1. Ask `gpt-3.5-turbo` whether the completion contained hallucination(s), using a detailed and carefully engineered prompt.
2. Run step 1 multiple times, typically 5. (We use batch inference here for its speed and cost advantages.)
3. Divide the number of "yes" answers from step 2 by the total number of answers to produce a score between 0 and 1.

Among metrics previously proposed in the literature, *ChainPoll* is perhaps closest to G-Eval [3].

However, we find that *ChainPoll* dramatically outperforms G-Eval across the entirety of *RealHall*. We attribute this to a number of key differences between *ChainPoll* and G-Eval:

- We put considerable effort into prompt engineering. In particular, we phrase our chain-of-thought prompt in a way that reliably elicits a very specific and systematic explanation from the model, an prompting approach we call "detailed CoT."
  - By contrast, the prompts used in [3] either did not use chain-of-thought, or asked for the answer before the chain-of-thought explanation, which prevents the answer from leveraging the reasoning in the explanation.
- We request boolean judgments, rather than numeric scores. In early experiments on this distinction, we observed that boolean judgments work better than scores, even when eliciting only a single completion.
- We use `gpt-3.5-turbo`, while [3] used either `text-davinci-003` or `gpt-4`[5].

4.2.1 *ChainPoll-Correctness* and *ChainPoll-Adherence*

Depending on the situation, a user may want to detect open-domain hallucinations, closed-domain hallucinations, or both.

We define a *ChainPoll*-based metric for each of these cases.

- *ChainPoll-Correctness* uses *ChainPoll* to detect open-domain hallucination.
- *ChainPoll-Adherence* uses *ChainPoll* to detect open-domain hallucination.

The two metrics differ only in the prompt format used when prompting `gpt-3.5-turbo`. In *ChainPoll-Correctness*, the prompt format asks the model to look for open-domain hallucinations; in *ChainPoll-Adherence*, the prompt format asks the model to look for closed-domain hallucinations.

4.2.2 Re-using chains of thought for explainability

In the *ChainPoll* approach, the LLM is asked to judge whether or not the original completion contained hallucination, justifying its answer with a chain-of-thought (CoT) explanation. While we use CoT primarily as a means to improve the quality of the model's final judgements, it also opens up interesting avenues for explaining that judgment to an end user.

That is, we can re-use the CoT text generated by the model as a *justification* for the judgment that the completion did, or did not, contain hallucination(s)[6].

We find that the chains of thought written by `gpt-3.5-turbo` are coherent and frequently convincing. An example is given below, explaining why a COVID-QA completion was hallucinated:

```
The response claims that the study described is a descriptive study of severe
hospitalized cases of laboratory-confirmed influenza during five epidemic seasons
(2010-2015).
```

[5]While GPT-4 performs very well at hallucination detection, we consider it too expensive for routine use in production. Our goal is here to offer high quality hallucination detection without the expense of GPT-4. (OpenAI's pricing is such that aggregating over multiple `gpt-3.5-turbo` completions is still much cheaper than generating a single completion with GPT-4.)

[6]An interesting line of recent work has called into question whether model-generated chains of thought faithfully reflect the model's actual reasoning process [16]. From our perspective, this does not reduce the value of model-generated chains of thought as user-facing *justifications* for model-generated judgments.
For example, suppose the model tells us that a particular completion was hallucinated, and it provides a CoT consistent with this judgment. *Whether or not* the CoT reflects how the model actually arrived at its judgment, it is nonetheless an argument that *could* be made in favor of that judgment. If the argument is correct, then it has value for the end user, even if the model arrived at its (correct) judgment through some other reasoning process.

```
To verify whether this claim is supported by the documents, we need to check if any
of the documents mention a study or research that matches the description.
The first document mentions the need to deepen knowledge of the effect of aging,
chronic diseases, and immune response in order to implement preventive measures.
This document does not directly state that it describes a study of severe
hospitalized cases of laboratory-confirmed influenza.
The second document does not provide any relevant information to support or refute
the claim.
The third and fourth documents both mention a study design and protocol for
investigating social network patterns and respiratory disease transmission and
control.  These documents do not mention severe cases of laboratory-confirmed
influenza.
Based on the analysis of the documents, none of them provide direct support for the
claim that the study described is a descriptive study of severe hospitalized cases
of laboratory-confirmed influenza during five epidemic seasons (2010-2015).
Therefore, the response is not supported by the documents.
```

5 Results

We present the results of our evaluations here, demonstrating that ***ChainPoll* is a new state-of-the-art.**

Across a diverse range of benchmark tasks, the *ChainPoll* outperforms all other methods – in most cases, by a huge margin.

Taking efficiency into account *ChainPoll*'s lead is even larger. It outperforms the next-best method, SelfCheck-BertScore, while using only 1/4 as much LLM inference, and without using an additional model like BERT.

Unlike all other methods considered here, *ChainPoll* also provides human-readable verbal *justifications* for the judgments it makes, via the chain-of-thought text produced during inference.

5.1 AUROC scores

Metric	Average AUROC
ChainPoll-Correctness	**0.772**
SelfCheck-Bertscore	0.670
SelfCheck-NGram	0.636
G-Eval	0.574
Max pseudo-entropy	0.565
GPTScore	0.489
Random Guessing	0.500

Table 3: Open-domain hallucination detection performance on *RealHall Open*, averaged across datasets.

Metric	Average AUROC
ChainPoll-Adherence	**0.789**
SelfCheck-Bertscore	0.675
SelfCheck-NGram	0.652
TRUE	0.593
G-Eval	0.584
Max pseudo-entropy	0.535
GPTScore	0.558
Random Guessing	0.500

Table 4: Closed-domain hallucination detection performance on *RealHall Closed*, averaged across datasets.

Per-dataset AUROC scores are provided in Appendix A.7.

5.2 Plots

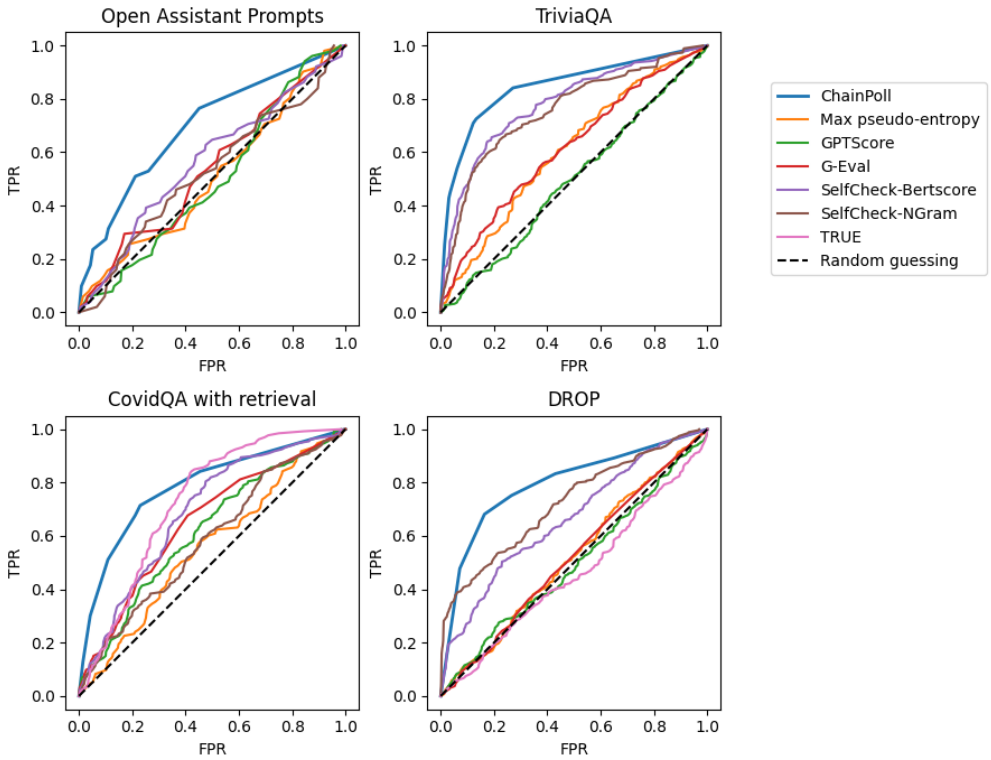

Figure 1: ROC curves for hallucination detection across *RealHall* datasets. The ChainPoll curves are *ChainPoll-Correctness* in the top row, and *Chainpoll-Adherence* in the bottom row.